What remains open in NIST prime-curve seed generation?

16 October 2026. Derived historical-process assessment of ten published and eleven reported passed-over qualifying blocks.

The ten published prime-field seeds are exact 20-byte blocks from six timestamp-initialized `lrand48` streams. The missing historical fact is why eleven other reported blocks in those streams, also yielding prime-order curves of the required size, were passed over. The original program might not have evaluated every window. The complete data and citations are in the Markdown report, with exact derived checks in process_checks.json.

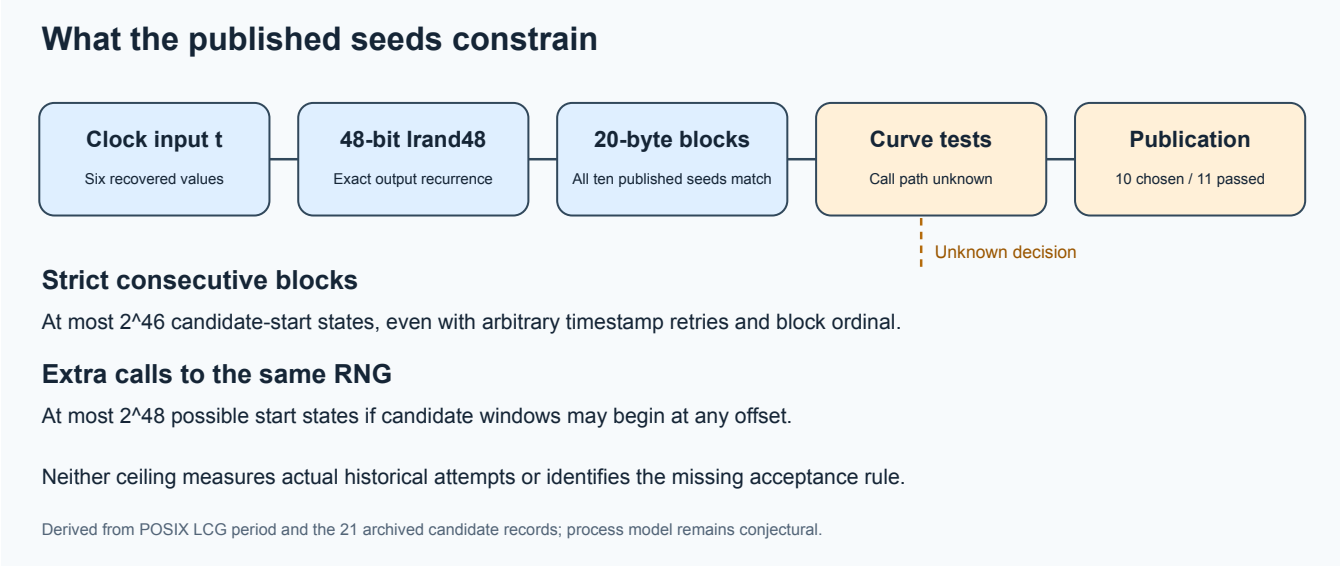


Figure 1: Known seed source, unknown acceptance step, and the two state-space ceilings.

What the new check rules out

For all 17 reported qualifying blocks with `p mod 4 = 3`, the recorded prime-order coefficient is the usual exponentiation square root of `-27/c`, including all eight passed-over blocks in those fields. The other four blocks are P-224: its two coefficient signs define isomorphic curves, so sign choice cannot disqualify its three passed-over blocks. All 21 satisfy the checked ANSI order-size and MOV screens. These results rule out a simple fixed-sign or documented-order-filter explanation for the reported omissions.

The earlier exact audit also found that a rule requiring positive trace, `n < p`, or a twist without small prime factors conflicts with published controls. A later factorization showed that passed-over P-224 block 1485 has a 122-bit largest twist prime factor, versus 118 bits for published P-224. Maximizing that twist metric cannot explain the choice.

How much generator choice remains?

The 48-bit `lrand48` recurrence has a full `2^48`-state cycle. `srand48(t)` fixes the low 16 state bits to `0x330E`. The first output state of each consecutive 20-draw block is 1 modulo 4, and moving to the next block advances 20 states. Since `gcd(20, 2^48)=4`, a complete strict-block stream visits exactly `2^46` possible first states; changing the timestamp changes the starting position within that set. Thus at most `2^46` distinct 20-byte seeds are available under the exact reconstructed block rule, irrespective of timestamp restarts. Extra calls **between** 20-byte windows allow any start state and raise the ceiling to `2^48`. Both ceilings assume consecutive draws **within** each candidate; changing that rule is a different model. They count possible generator states, not observed attempts.

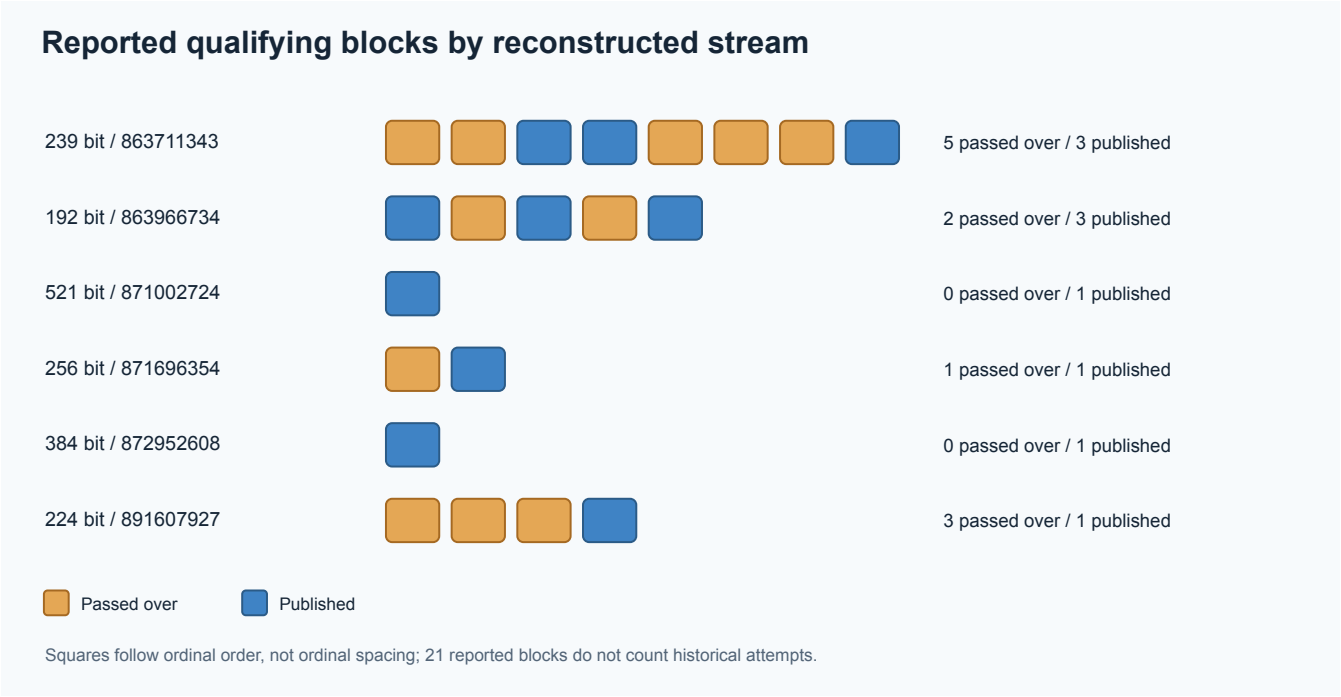


Figure 2: The 21 reported qualifying blocks; orange blocks were passed over.

Remaining process models

Extra generator calls could bypass otherwise qualifying windows, but must reproduce the observed 20-draw alignment of published seeds. A historical point-count or certificate implementation could reject curves that modern arithmetic verifies, but no such failure has been reproduced. An additional curve-quality test or manual choice is logically possible; it needs a rule specified before testing an independent historical decision. Arbitrary timestamp restarts are also uncounted. The six May 1997 published base points were reconstructed using a separate FreeLIP generator, weakening one specific same-stream consumption explanation for those six.

The fastest decisive evidence is a provenanced copy of Bob Reiter's program, call log, or contemporary output files. A second route is a frozen process model that predicts independent historical selections. Neither the reported 21 labels nor the `2^46` ceiling establishes how many candidates were actually inspected, a hidden mathematical weakness, or a universal security conclusion.

Sources: Anthropic's reconstruction Valsorda's candidate list text copy of ANSI X9.62-1998 FIPS 186-2 prior internal research and source hashes in the full report.